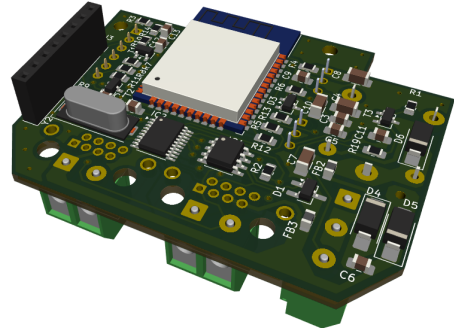


1 Features

- Control of an output actuator using NFC tags
- Access rights evaluation
- Designed for a 24V actuator
- Flush-mounted installation in an electrical box
- Bus connection via RJ45 or screw terminals
- Tag reading indication



2 Device Description

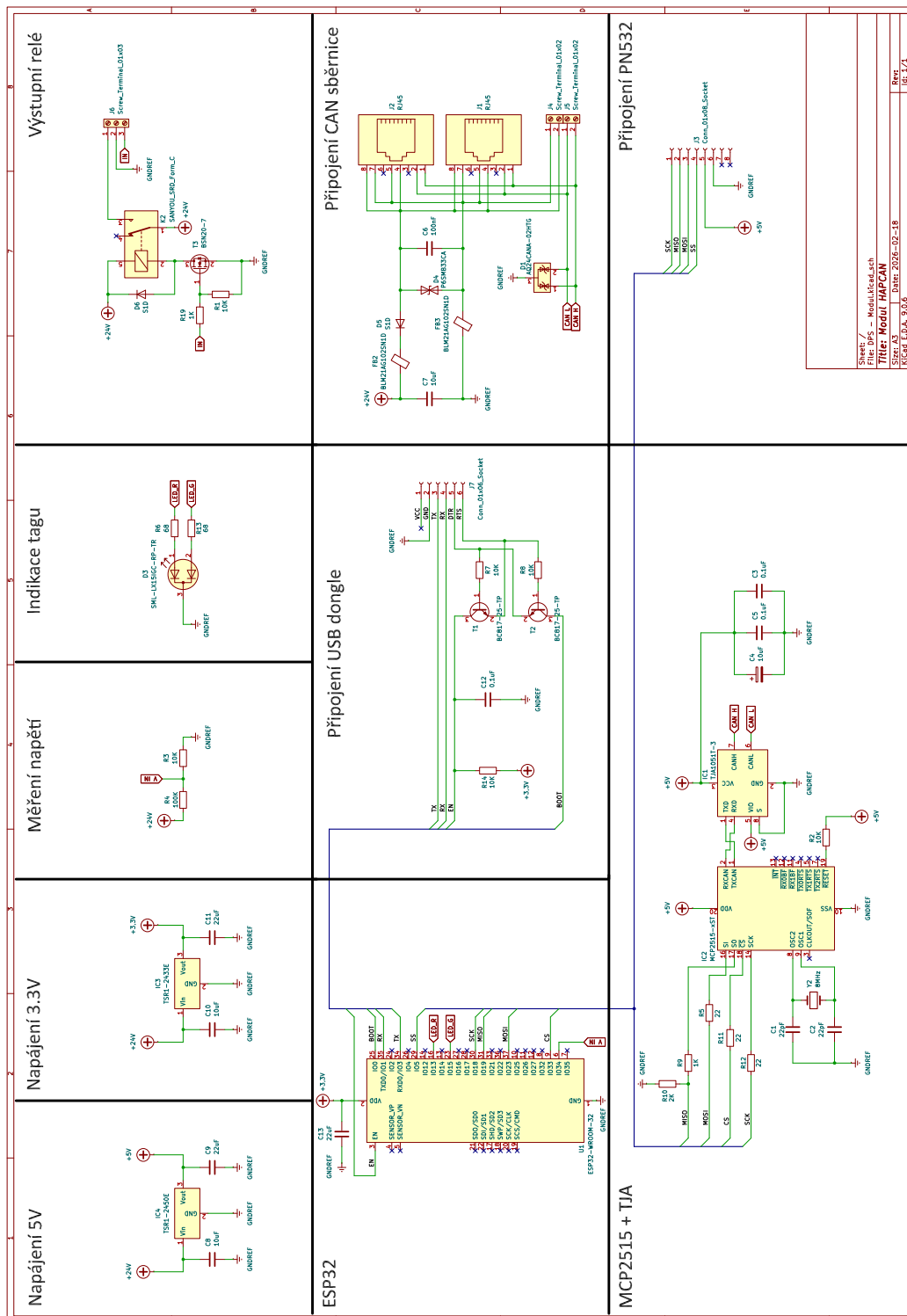
This universal module serves as an end device for contactless access control and output actuator management (e.g., an electrical lock). The device is designed with an emphasis on full compatibility with the HAPCAN protocol, allowing for easy and reliable integration into this system. The device functions exclusively as a peripheral that facilitates tag reading and the physical triggering of the lock, while the actual verification of access rights can be configured in the HAPCAN Programmer application. After the physical reading of the identification tag, the data is sent over the bus for further processing within the system. The module triggers the output device only upon receiving the appropriate control message.

2.1 Complete Documentation

Complete manufacturing documentation for the hardware, firmware source codes, and 3D printing data for the protective cover are freely available on the GitHub platform at the following link: https://github.com/Poky88/HAPCAN_Modul_RFID_CTECKA

3 Hardware

3.1 Schematic Diagram



3.2 Bill of Materials

Reference	Component	Value	Quantity	Footprint
IC4	DC/DC Converter	5V	1	3-SIP module
IC3	DC/DC Converter	3,3V	1	3-SIP module
R4	SMD Resistor	100k	1	0805
R1,R2,R3,R7,R8,R14	SMD Resistor	10k	6	0805
R10	SMD Resistor	2k	1	0805
D5, D6	Rectifier Diode	-	2	DO-214AC, SMA
K2	Relay	24V	1	SRD, C
R9, R19	SMD Resistor	1k	2	0805
T3	MOSFET Transistor	-	1	SOT-23
U1	ESP32-WROOM-32E	-	1	38-SMD module
T1, T2	NPN Transistor	-	2	SOT-23
C3, C5, C6, C12	SMD Capacitor	0,1uF	4	0805
FB2, FB3	Ferrite Bead	-	2	0805
D4	Protection Diode	-	1	DO-214AC, SMA
D1	Protection Diode	-	1	SOT-23
R5, R11, R12	SMD Resistor	22	3	0805
C1, C2	SMD Capacitor	22pF	2	0805
Y2	Crystal	8MHz	1	HC-49
IC2	MCP2515T-I/ST	-	1	20-TSSOP
IC1	TJA1051T/3/2Z	-	1	8-SOIC
C4	SMD Capacitor	10uF	1	1206
(C7), C8, C10	SMD Capacitor	10uF	2-3	1206
C9, C11	SMD Capacitor	22uF	2	0805
D3	Indicator LED	-	1	SOT-23
R6, R13	SMD Resistor	68	2	0805
J4, J5	Terminal Block	-	2	10.5x7.8, 5mm raster
J6	Terminal Block	-	1	15.5x7.8, 5mm raster
J3	Pin Header	-	1	1x08, 2.54mm raster
J7	Pin Header	-	1	1x06, 2.54mm raster
J1, J2	RJ45 Connector	-	2	RJHSE5380

Table 1: Bill of Materials

3.3 Printed Circuit Board - Assembly Plan



Figure 1: Assembly plan - TOP

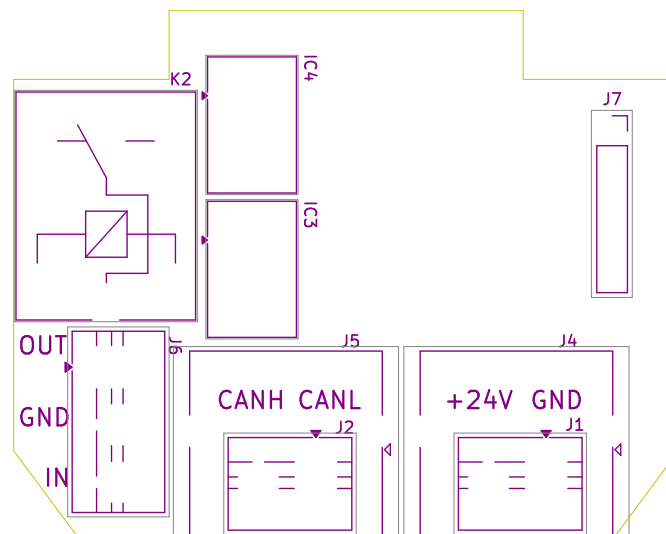


Figure 2: Assembly plan - BOT

4 Typical Wiring

The following diagram shows a wiring example with a general actuator (e.g., an electrical lock). The user can choose any output actuator. For the correct function of the module, it is essential to connect it to the CAN communication bus. The diagram also illustrates the possibility of bringing an external control signal to the **IN** terminal to control the output device. For maximum clarity, all connection terminals are clearly marked.

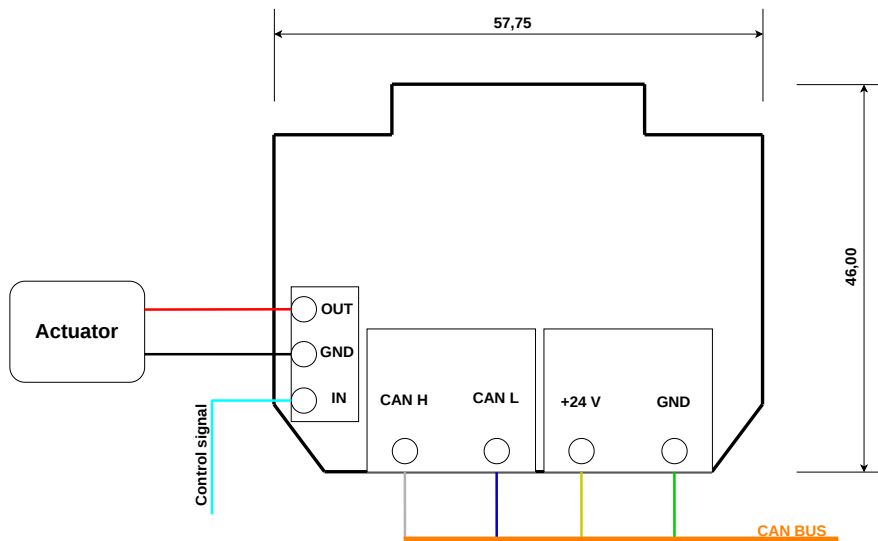


Figure 3: Typical wiring

4.1 Installation into an Electrical Box

The device is first placed into a protective cover, which, in addition to mechanical protection, also ensures the correct fixation and positioning of the reader. To successfully install the entire module into a flush-mounted electrical box, some of its internal plastic parts must be removed. The extent of these modifications always depends on the specific type of box used. The following figures show the progress of these modifications and their final result.

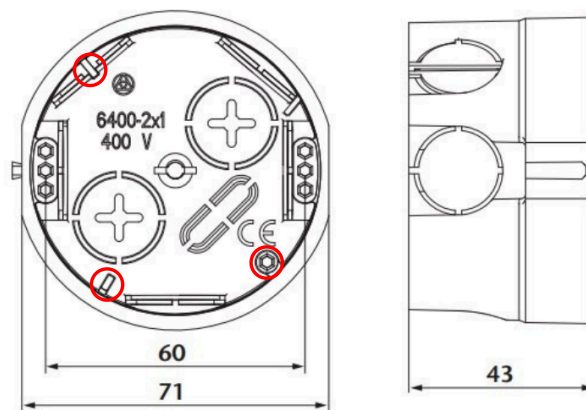
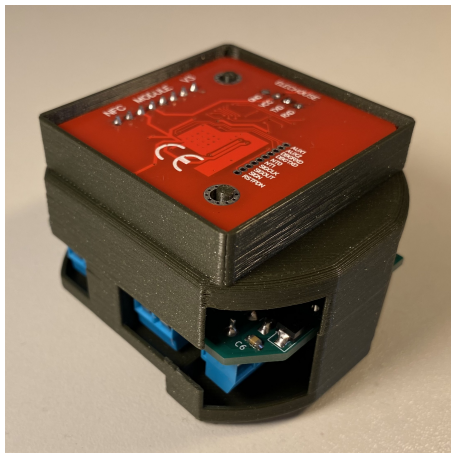


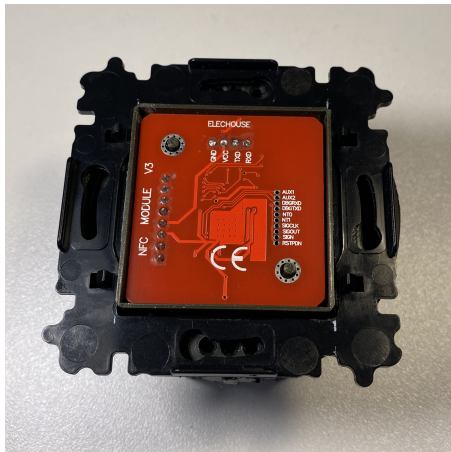
Figure 4: Removal of obstructing parts



(a) Fully assembled module



(b) Module placed in the box



(c) Grid on the box



(d) Complete installation

Figure 5: Module installation into an electrical box

5 License



HAPCAN is a trademark of its respective owners. This project is an independent implementation compatible with the HAPCAN system.

This hardware is free hardware: you can redistribute it and/or modify it under the terms of the GNU General Public License as published by the Free Software Foundation, either version 3 of the License, or (at your option) any later version.

This hardware is distributed in the hope that it will be useful, but WITHOUT ANY WARRANTY; without even the implied warranty of MERCHANTABILITY or FITNESS FOR A PARTICULAR PURPOSE. See the GNU General Public License for more details.

You should have received a copy of the GNU General Public License along with this program. If not, see <http://www.gnu.org/licenses/gpl-3.0.html>.